DreamZero: World Action Models are Zero-shot Policies

Ye, Ge, Zheng, Gao, Yu et al. (NVIDIA, 2026)

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WoRV / MaumAI

The Second Pre-training Paradigm

1st Paradigm: Next-word prediction → Language intelligence (LLMs)

2nd Paradigm: **Next physical state prediction** → Embodied intelligence (World Models)

> *"2026 will go down as the first year that Large World Models lay real foundations for robotics.*

— Jim Fan, *"The Second Pre-training Paradigm"* (Fan, 2026)

Why Not Just Use VLMs for Robots?

- VLMs are **language-first** — vision is a "second-class citizen"
- Most VLM parameters encode *knowledge*, not *physics*
- Static image–text pretraining $\neq$ spatiotemporal understanding

Commentary: *"Language is a bottleneck, a scaffold, not a foundation" for physical intelligence.*

The Data Problem in Robotics (1/2)

How much data does each domain have?

- **LLMs:** text from **billions of people** across history
- **Vision:** images from **billions of cameras** over decades
- **Robotics:** **thousands of hours** of teleoperation $\rightarrow$ **massive gap**

In-the-wild vs. On-demand Data

Text and images are "in-the-wild" — generated continuously.
Robot data is "on-demand" — requires expensive teleoperation.

The Data Problem in Robotics (2/2)

> *“You can’t teleoperate your way to the long tail.*

— Joel Jang (Jang, 2026)

“Intelligence is a function of experience.”

$\Rightarrow$ Robotics needs **vastly more experience** to match human-level physical intelligence.

The Rise of VLAs and Their Limitations

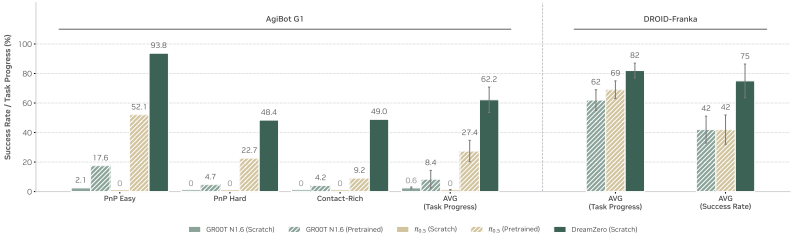
Vision-Language-Action Models (VLAs):

- RT-2, OpenVLA, GR00T, π_0 (Brohan et al., 2023; Kim et al., 2024; Bjorck et al., 2025; Black et al., 2024)
- Extend VLMs $\rightarrow$ predict motor actions
- **Strong semantic understanding**: "pick up the red cup"

But:

- Trained on **static images** $\rightarrow$ lack spatiotemporal priors
- Fail on novel *motions*: "untie the shoelace"
- Scaling model size alone **does not help**

VLAs: The Evidence



VLA from-scratch at ∼0% on diverse data

Commentary: *The fundamental gap — VLMs encode "**what** to do" but not "**how** to move".*

Key Insight: Video as Dense Physical Representation

Core Idea

Video diffusion models trained on web-scale data encode:
spatial geometry · **object dynamics** · **contact physics** · **motion patterns**

- Every consecutive frame pair $\rightarrow$ **dense supervision** (vs. sparse task-level labels in VLAs)
- "Generating pixels is the only way for humans to *verify* that the model truly understands" — Joel Jang (Jang, 2026)

The Paradigm Shift

> *"Apes may not have good language models, but they surely have a robust mental picture of counterfactuals.*

— Jim Fan (Fan, 2026)

Commentary: *This is the core paradigm shift — from discrete task policies to* ***continuous world modeling****.*

World Action Models: The Big Picture

WAM = Video Prediction $\times$ Inverse Dynamics Model

$$\underbrace{P(\mathbf{O}, \mathbf{A} \mid \text{ctx})}_{\text{DreamZero}} = \underbrace{P(\mathbf{O} \mid \text{ctx})}_{\text{video pred.}} \times \underbrace{P(\mathbf{A} \mid \mathbf{O}, \text{ctx})}_{\text{IDM}}$$

IDM (Inverse Dynamics Model): observes a sequence of states and infers what actions caused those transitions — watching video and reverse-engineering motor commands.

- Instead of two separate models $\rightarrow$ **single end-to-end model**
- **DreamZero:** 14B autoregressive WAM built on **Wan2.1** video backbone (Team Wan, 2025)

DreamZero: Headline Results

Key Achievements

- **2× improvement** over state-of-the-art VLAs
- **38× inference speedup** → real-time at 7Hz
- **Cross-embodiment transfer** from video-only data

The Human Data Scaling Pathway

> *Humans are robots that have already been deployed at scale.*

— Joel Jang (Jang, 2026)

The math:

- 8 billion "units" $\times$ 16 waking hrs/day $=$ 128B person-hours *per day*
- Even capturing 0.1% $\rightarrow \approx$ 100M hours of usable video
- $\approx$ 150 human lifetimes of experience

The vision: "Train once on human experience, deploy everywhere with calibration"

Human Data: Early Signal

Cross-Embodiment Transfer

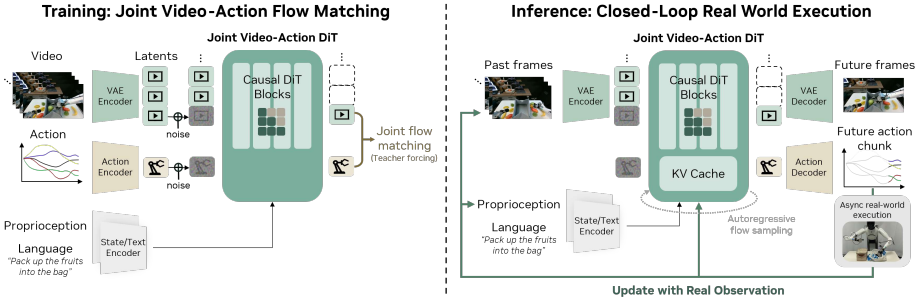
12 min of human egocentric video
$\rightarrow$ $>$**42% relative improvement** on unseen robot tasks

Commentary: *This is the sleeper insight — human video as the ultimate robot training data.*

How Does DreamZero Work?

Architecture & Training

Architecture Overview



Inputs:
Visual context (VAE)
Language (text encoder)
Proprioceptive state

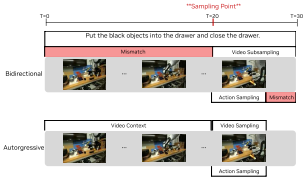
Backbone:
Autoregressive DiT
(Diffusion Transformer)
with flow matching

Outputs:
Joint video frames
+ action chunks

Why Autoregressive? AR vs. Bidirectional

Bidirectional problems:

- Fixed-length sequences → **video subsampling**
- Distorts native FPS → breaks alignment
- Especially harmful mid-task



Autoregressive advantages:

- Preserves **native frame rate**
- **KV-cache** for 3–4× speedup
- Natural modality alignment
- Variable-length training

Commentary: *AR wins on both quality AND speed — KV-cache enables 3–4× speedup while preserving the temporal signal the IDM needs.*

Chunk-wise Generation

Chunking Strategy

- Video: $K = 2$ **latent frames** per chunk
- Action horizon: $H = 48$ (AgiBot, 30Hz) or $H = 24$ (DROID, 15Hz)
- Each chunk $\approx$ **1.6 seconds**
- Max 4 chunks $\rightarrow$ **6.6 seconds** context

Key design choices:

- Variable-length training (like LLM token sequences)
- Video at 5 FPS, actions at 30Hz (AgiBot)
- 33 raw frames $\rightarrow$ 8 latent frames (4×2)

Flow Matching: Intuition

What is Flow Matching?

Learn a **velocity field** that transports noise to data along straight paths — simpler and faster than standard diffusion.

Analogy: Think of a GPS that plots a straight-line route from "lost" (noise) to "home" (data). Flow matching learns that route; standard diffusion takes a random walk.

- At $t = 0$: pure noise. At $t = 1$: clean data.
- The model predicts the **velocity** pointing from noise to data.
- **Shared timestep** t_k for video and action $\rightarrow$ aligned convergence.

Flow Matching: Equations

Noisy interpolation:

$$\mathbf{z}_{t_k}^k = t_k \mathbf{z}_1^k + (1 - t_k)\mathbf{z}_0^k, \quad \mathbf{a}_{t_k}^k = t_k \mathbf{a}_1^k + (1 - t_k)\mathbf{a}_0^k$$

where $\mathbf{z}_0^k, \mathbf{a}_0^k \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $t_k \in [0, 1]$.

Velocity prediction loss:

$$\mathcal{L}(\theta) = \mathbb{E}\left[\frac{1}{K}\sum_{k=1}^{K} w(t_k)\left\|\mathbf{u}_\theta(\cdots) - \mathbf{v}^k\right\|^2\right]$$

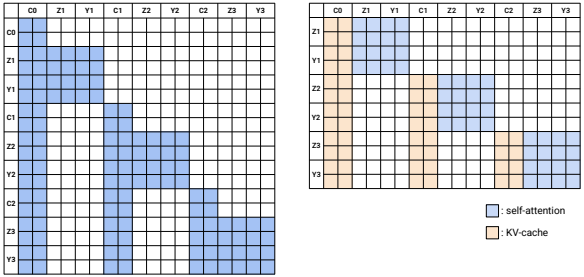
Flow Matching: Notation Guide

Key Symbols:

- $\mathbf{u}_\theta$: the DiT model (predicts velocity)
- $\mathcal{C}_k$: clean context from previous chunks
- $\mathbf{c}$: language instruction
- $\mathbf{q}_k$: proprioceptive state
- $w(t_k)$: predefined weight function

Teacher forcing: denoise noisy current chunk conditioned on **clean** previous chunks (like LLM training with ground-truth tokens).

Attention Masking Strategy



(a) Attention mask (Training)

(b) Attention mask (Inference)

Key Rules

- **Training:** noisy chunk attends ONLY to **clean context**
- **No cross-attention** between noisy video and noisy actions within same chunk

Inference: Ground-Truth Injection

The Elegant Trick

1. Denoise one chunk at a time (autoregressive)
2. Execute action chunk on robot $\rightarrow$ get **real observation**
3. **Replace predicted frame with ground-truth** in KV cache
4. This **eliminates error accumulation!**

Why this is unique to WAMs:

- Pure video generation: errors compound
- WAMs operate in **closed-loop** — reality corrects the model
- WAMs **self-correct via reality** — no other paradigm gets this for free

DreamZero-Flash: Faster Inference

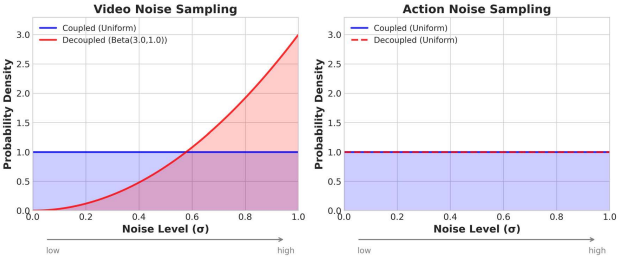
Problem: 4 denoising steps = 350ms (too slow for some applications)

Solution: **Decoupled noise scheduling**

- Video: $t_{\text{vid}} = 1 - \eta, \eta \sim \text{Beta}(7, 1)$
  $\rightarrow$ biased toward **high-noise** (mean $t \approx 0.125$)
- Action: $t_{\text{act}} \sim \text{Uniform}[0, 1] \rightarrow$ standard

Trains actions to denoise from *predominantly noisy* visual context.

DreamZero-Flash: Results



Result

1-step: **74%** task progress (only 9% drop from 4-step at 83%)

38× Inference Speedup Breakdown

Optimization	H100	GB200
Baseline	1×	1.1×
System-level		
+ CFG Parallelism	1.9×	1.8×
+ DiT Caching	5.5×	5.4×
Implementation-level		
+ torch.compile	8.9×	10.9×
+ Kernel & Scheduler	9.6×	14.8×
+ Quantization	—	16.6×
Model-level		
+ DreamZero-Flash	—	38×

Final: 14B at **7Hz** on 2×GB200 (150ms per chunk)

Data Philosophy: Diversity Over Repetition

Counterintuitive Finding

Same 500 hours of data:
Diverse (22 envs, long-tail skills): **50% task progress**
Repetitive (70 tasks, many reps): **33% task progress**
$\rightarrow$ **+51% relative improvement** from diversity

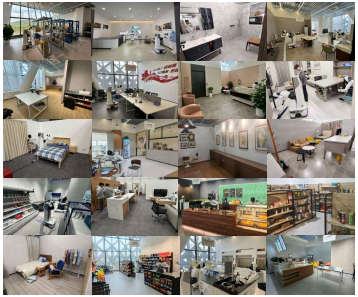
Why?

- Video prediction inherited from pretraining
- Robust IDM requires **diverse state-action correspondences**
- Repetitive data $\rightarrow$ narrow IDM

AgiBot G1 Dataset

Data:

- **500 hours** teleoperation
- 7.2K episodes
- **22** real-world environments
- Avg episode: 4.4 min, $\sim$42 subtasks

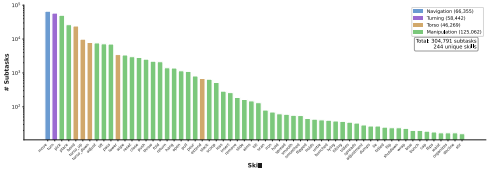


22 diverse environments

Training Configuration

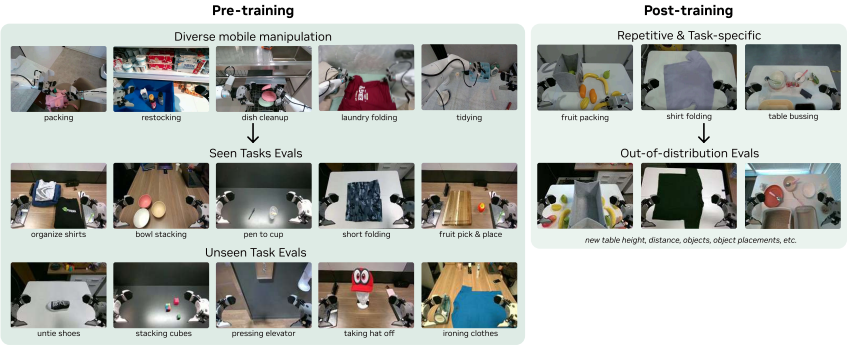
Training:

- Backbone: **Wan2.1-I2V-14B**
- 100K steps, batch size 128
- **Freeze:** text enc, image enc, VAE
- **Update:** DiT + state/action enc/dec
- Post-training: 50K steps per task



Skill distribution

Evaluation Protocol: Setup



Default: **unseen environments**, **unseen objects**
Training and evaluation in **different geographic locations**

Evaluation Protocol: Details

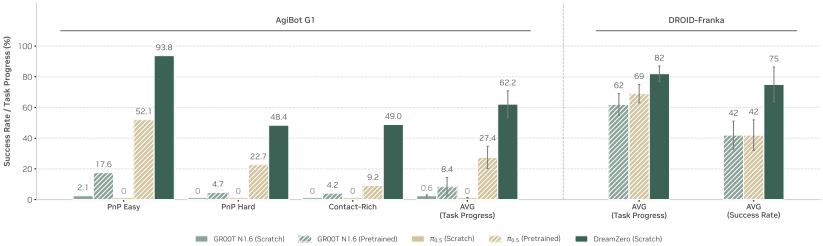
True Out-of-Distribution

- Seen tasks: 10 tasks $\times$ 8 rollouts $\times$ 4 robots = 80 rollouts
- Unseen tasks: 10 tasks absent from training
- Baselines: GR00T N1, $\pi_{0.5}$ (from-scratch & from-pretrained)

Does It Actually Work?

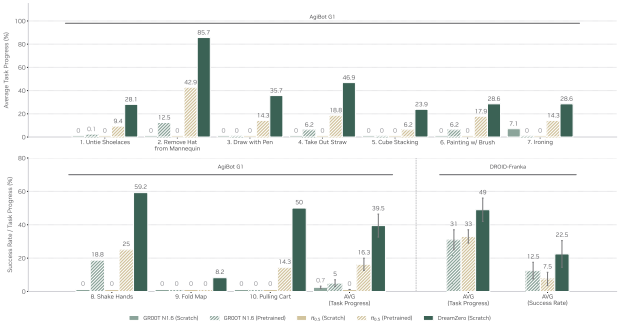
Experimental Results

Seen Tasks: 2× Improvement



DreamZero: **62.2%** avg. task progress
Best VLA (pretrained): 27.4% → **2.3**×
VLA from-scratch: ∼**0%**

Unseen Tasks: Zero-shot Generalization



DreamZero: **39.5%** on novel tasks
Best VLA (pretrained): 16.3% $\rightarrow$ **2.4**$\times$

Unseen Tasks: Standout Results

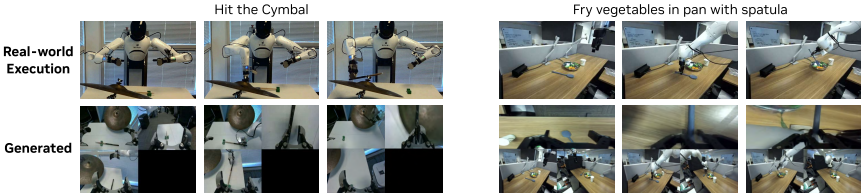
Standout tasks:

- Remove hat: **85.7%**
- Shake hands: **59.2%**

VLAs overfit to pick-and-place regardless of instruction.

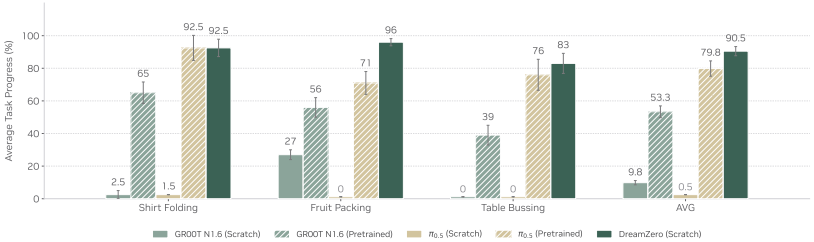
> ***Commentary:*** *The key result — DreamZero does things it was* ***NEVER trained to do****.*

Joint Video-Action Prediction in Action



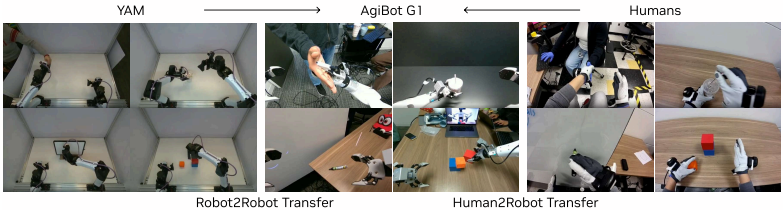
- Tight alignment between predicted and actual trajectories
- Most failures: video generation errors, not action extraction

Post-training: Retained Generalization



- **Matches or outperforms** pretrained VLAs
- Evaluated in **unseen environments** even after task-specific training

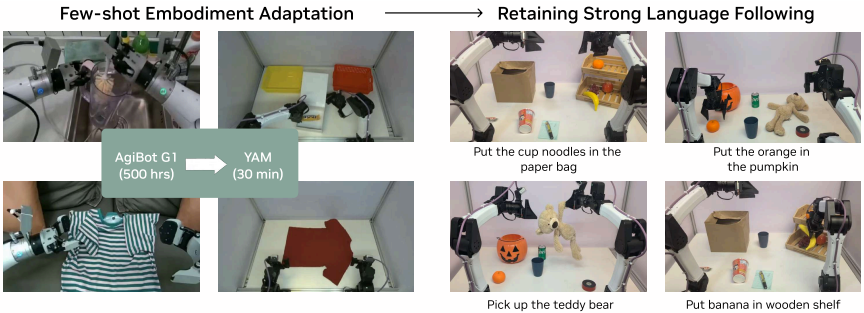
Cross-Embodiment Transfer



Method	Task Progress
DreamZero (baseline)	38.3%
+ Robot2Robot (YAM, 20 min)	**55.4%** (+44.6% rel.)
+ Human2Robot (12 min)	**54.3%** (+41.8% rel.)

Commentary: *Video-only data (no action labels!) opens a* ***massive scaling pathway*** *— YouTube hosts* $>$***800M hours*** *of activity video.*

Few-shot Adaptation to New Robot



- 55 trajectories (∼30 min play data) on YAM robot
- Retains **language following** + generalizes to novel objects

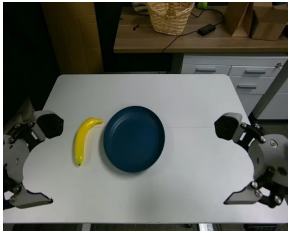
Free-form Evaluation

100+ novel tasks via natural language instructions

“Pop the balloon”, “Press elevator button”, “Paint”, “Iron clothes”

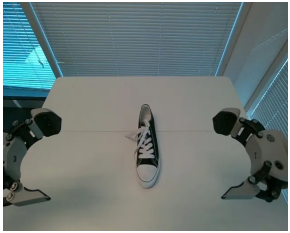
Task Diversity: 103 Tasks Across 22 Environments

Seen Tasks (10 evaluated):



take, place, push, open, pour

Unseen Tasks (10 evaluated):



iron, untie, shake hands, remove hat

Full video gallery: `dreamzero0.github.io`

Ablation: What Matters Most?

Arch	Size	Data	Progress
Q1. Data Diversity			
DZ (AR)	14B	Repetitive	33%
DZ (AR)	14B	Diverse	**50%**
Q2. Model Scale			
DZ (AR)	5B	Diverse	21%
DZ (AR)	14B	Diverse	**50%**
VLA	5B	Diverse	**0%**
VLA	14B	Diverse	**0%**
Q3. Architecture			
DZ (BD)	14B	Diverse	50%
DZ (AR)	14B	Diverse	**50%**

All on PnP Easy, 50K steps, batch 32. DZ = DreamZero.

Why VLAs Fail While WAMs Succeed

VLAs:

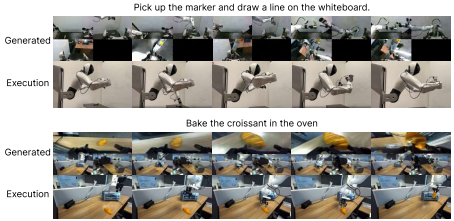
- **Static images** $\rightarrow$ no temporal dynamics
- Scaling does not help: **0% at 5B and 14B**
- Repetitive data + static priors = **overfitting**

WAMs:

- **Video** $\rightarrow$ rich temporal dynamics
- Scaling improves video quality $\rightarrow$ **better actions**
- Diverse data + video priors = **generalization**

Commentary: *A **fundamental architectural difference**, not data/compute. VLMs encode semantics; video models encode physics. For robotics, physics wins.*

Failure Analysis: When DreamZero Breaks



Generated vs. executed trajectories

Error causal chain:

1. Video backbone hallucinates
2. Frames show wrong trajectory
3. IDM extracts actions from wrong video
4. Robot executes incorrect motion

Commentary: *The tight coupling is both strength and weakness —* ***video quality is the bottleneck****.*

Under the Hood

Code-Level Insights

Code: Multi-Embodiment Architecture

Key Code Patterns

- `CategorySpecificLinear`: separate weights per embodiment
- `MultiEmbodimentActionEncoder`: category-specific MLPs
- Supports up to **32 embodiments**
- Relative action normalization

Commentary (from code): *Each robot gets its own "translator" while sharing the world model.*

Code: Making 14B Real-Time

System Engineering

- **VRAM:** dynamic GPU–CPU offloading
- **FP8 inference:** automatic scaling
- **WebSocket:** distributed multi-GPU
- **DiT caching:** skip converged passes

Quantization stack:

- Weights + activations: NVFP4
- QKV, Softmax: FP8
- Non-linear ops: FP16

880 tokens per frame sequence
(33 frames $\times$ $\sim$27 tokens/frame)

Commentary (from code): *Far beyond typical research code — production-grade systems engineering.*

Code: Data Pipeline

Data Infrastructure

- **LeRobot v2 format** with GEAR metadata
- Multi-view camera alignment
- Sharded dataset loading (10% per shard)
- Action filtering: remove idle actions

Commentary (from code): *Data infrastructure is as important as the model architecture.*

Code: WAM as Modified Video DiT

Core Architecture

- Core file: `wan_video_dit_action_causal_chunk.py`
- Action head integrated **INTO** diffusion process (not a separate MLP)
- Flow matching scheduler + **UniPC multi-step** for Flash mode
- Codebase: `groot/vla/model/dreamzero/`

Commentary (from code): *DreamZero is not "video model + action decoder" — it is a* ***single DiT*** *that jointly denoises both modalities.*

What Does It All Mean?

Discussion & Takeaways

Limitations

Paper Limitations:

1. No established **scaling laws**
2. Context limited to **6.6s**
3. Still **7Hz** vs. 20Hz+ VLAs
4. **Sub-cm precision** gap
5. Separate training per embodiment

Practical Constraints:

- $2\times$GB200 for real-time control
- Not feasible on consumer hardware
- 14B params = significant VRAM

Long-horizon:

- 6.6s context $\rightarrow$ System 1 only
- Needs System 2 planner

The Unsolved Inverse Dynamics Problem

Adaptation is not free.

The credit card analogy:

- Dexterous hands: slide + pinch
- Grippers: push from edge
- *Same task, different motions*

IDM complexity scales with DOF:

- Simple grippers: relatively easy
- Dexterous hands: much harder
- Full humanoids: **hardest**

The Hardware vs. Algorithm Race

Hardware Path:
If humanoid actuators mature quickly → embodiment matching removes transfer loss

Algorithm Path:
If embodiment adaptation improves faster → intermediate morphologies remain viable

> *"We are back to the age of research — fundamentals matter again.*

— Jim Fan (Fan, 2026)

Both paths converge on the same data: **human video**

Critical Assessment

Strengths:

- ✓ Comprehensive evaluation protocol
- ✓ Honest failure analysis
- ✓ Code and weights released
- ✓ Multiple embodiments tested

Missing:

- Comparison with other WAMs (Cosmos Policy, UVAM)
- "From-pretrained" baseline used *different data*
- Multi-embodiment *unified* training
- GB200 dependency unclear

Commentary: *Current zero-shot is "AI Slop" phase (Jang, 2026) — correct direction, but* ***not deployment-ready*** *yet.*

What DreamZero Means for the Field

Personal Assessment

1. WAMs represent a **paradigm shift**: from “policy learning” to “world understanding”
2. Jim Fan’s “second pre-training paradigm” is happening **NOW**
3. “Diversity $>$ repetition” could **reshape robot data collection**
4. Cross-embodiment transfer via video-only data is the **sleeper result**
5. Video backbone quality = policy quality $\rightarrow$ “chain of thought in **visual space** rather than language space”

Key Takeaways

1. **WAMs jointly predict video + action** $\rightarrow$ better generalization
2. **Diverse data + video priors** = zero-shot robot skills
3. **AR + teacher forcing + GT injection** = elegant inference
4. **38$\times$ speedup** makes 14B practical
5. **Cross-embodiment transfer** via video-only data
6. **"Train once on human experience, deploy everywhere"**

The future of robotics is video-native, not language-native.

Thank You & References

Paper & Code:

- Ye et al. (2026)
- `dreamzero0.github.io`

Blog Posts:

- Jang (2026)
- Fan (2026)

Key References:

- Wan2.1 (Team Wan, 2025)
- GR00T N1 (Bjorck et al., 2025)
- $\pi_0/\pi_{0.5}$ (Black et al., 2024; Physical Intelligence, 2025)
- Flow matching (Liu et al., 2022; Lipman et al., 2022)
- DROID (Khazatsky et al., 2024)

Q&A

References I

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